STA2053: Special Topics in Applied Statistics

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Week 1: Introduction

Overview

- Introductions
- Course outline and goals
- Tools
- GLMs review (start)
- Lab: Git, tidyverse, R Markdown/Quarto

Introductions

Instructor

- Monica Alexander
- Email: monicaalexander@utoronto.ca.

Course outline and goals

Web

- Github: https://github.com/MJAlexander/sta2053
  - slides, labs, and data will be put here
- Quercus page
  - go here to submit research paper report and presentation (not labs)
  - class announcements will be put here

Course outline

- Topics will include generalized linear models, Bayesian inference, hierarchical models, generalized additive models involving non-parametric smoothing, temporal models, model evaluation and selection. We will also cover some core statistical computing techniques.
- A large focus of the outcomes on this course will also be on reproducible research, identifying and dealing with data and modeling issues, and model interpretation and communication.
- The focus in terms of methods is advanced regression techniques, fit using Bayesian inference. The focus in terms of coding/computation is becoming more comfortable and adept at efficient, reproducible coding and workflows (data, analysis, reporting and communicating results)

Course outline

- Throughout the course we will be using R in all examples
- Each week will be a lecture (~1.5hrs) then a lab

Assessment

- Lab exercises, 5 in total, 2% each
  - **Hand in via git**
  - Practice of concepts covered in the lecture
- Mid-term, 25%
  - After reading week
  - Assesses week 1-6
  - Short answer an multiple choice
- Final exam, 40%
- Research paper report and presentation 25%
  - I will assign you a paper to read, understand, report on and present
  - Report (10 %)
  - Presentation last week of class (15%)

Expectations

We will be doing applied statistics in the truest sense of the term

- Understand main ideas behind important techniques for applied statistics
- Coding in R (and in particular, the tidyverse, ggplot)
- Dealing with real data!
- R markdown or Quarto
- Git (terminal or desktop, not direct file upload)
- Code readability
- Clear communication of methods, findings, limitations
  - Data exploration is part of this!
- Aim for reproducible research

Course roadmap

Planned lecture content:

- Generalized linear models recap
- Bayesian inference
- Visualizing the Bayesian workflow and model checks
- Multilevel models
- Non-linear/ non-parametric models (Penalized splines)
- Temporal models / dealing with correlation

Roadmap

Planned lab content:

- Rmarkdown/Quarto, git
- Tidyverse
- EDA, data viz
- RShiny
- Stan
- Probably: web scraping
- Maybe: Extracting data from API (e.g. Facebook Spotify/Genius)

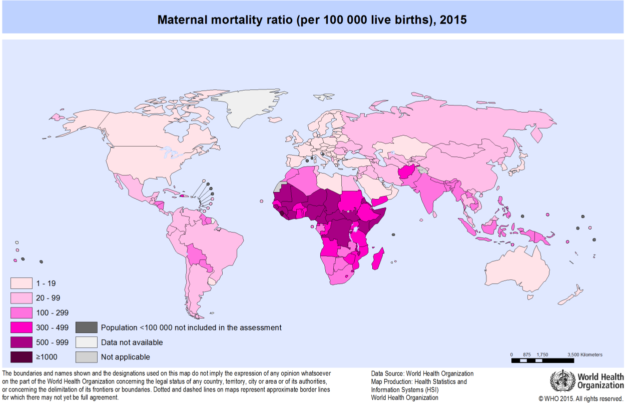
Motivating example

Global estimation of the causes of maternal death

- **Maternal mortality**: the death of a woman while pregnant or within 42 days of termination of pregnancy, from any cause related to or aggravated by the pregnancy.
- Very important indicator of health and development of a country
- Part of the Sustainable Development Goals (3.1)

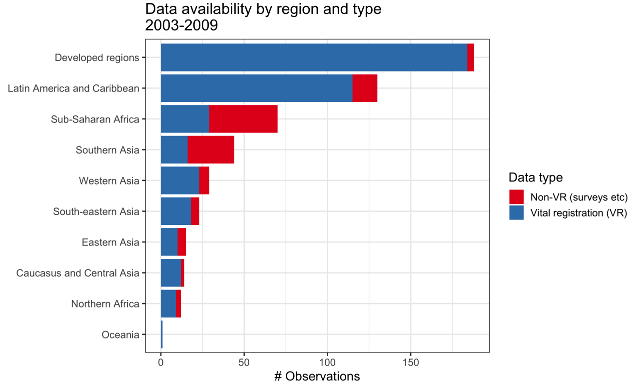
Global estimation of the causes of maternal death

- Large variation in maternal mortality ratio (deaths per 100,000 births) across the world (highest: 1150; lowest: 2)
- In order to reduce number of deaths, need to know underlying causes
- But this is difficult information to obtain/estimate



How do we get information on causes of (maternal) death?

- In high-income countries and some middle-income countries: civil registration systems
- In low-income countries: ???
  - surveys (why is this hard?)
  - facility-based administrative data
  - other specialized studies



How do we get information on causes of (maternal) death?

- If we had complete coverage of all deaths and a reliable way of classifying cause of death, then we could just count deaths and call it a day
- But in most countries (particularly high-burden countries) we have very little information, and what we do have is full of problems
- $\rightarrow$ Use statistical methods to obtain as reliable estimates as possible

Issues

To name a few:

- Years with no data
- Only some causes observed (even in high-income countries)
- Non-representative data (subnational, facility-based)
- Cause of death classification issues (death not witnessed, definition changes, differences across countries etc)
- Under/over-reporting (especially abortion)
- Not all civil registration systems are high quality
- Low death counts ($\sim$ 25 deaths in Australia)

Intro to statistical set-up

Notation:

- observations $i = 1, \ldots, n$
- d_i is total number of maternal deaths for the ith observation
- observed maternal deaths $\mathbf{y_i} = (y_{i,1}, \ldots, y_{i,7})$
- $y_{i,j}$ is the number of deaths due to cause j for the ith observation
- cause groups $j = 1, \ldots, 7$ corresponding to {ABO, EMB, HEM, SEP, DIR, IND, HYP}

Intro to statistical set-up

Think of deaths as a stochastic process:

- Given total number of maternal deaths d_i, the probability of a death is due to cause j is $p_{i,j}$ This is a Multinomial distribution, with 7 categories:

$$\mathbf{y_i} \sim \text{Multinomial}(d_i, \mathbf{p_i})$$
$$\mathbf{p_i} = (p_{i,1}, \ldots, p_{i,7})$$

- We observe $y_{i,j}$ and d_i
- We are interested in estimating $\mathbf{p_i}$. These will help us get estimates for the 'true' proportions $\mathbf{p_c}$ for countries $c = 1, \ldots, 193$ (UN member countries)

Intro to statistical set-up

$$\mathbf{y_i} \sim \text{Multinomial}(d_i, \mathbf{p_i})$$
$$\mathbf{p_i} = (p_{i,1}, \ldots, p_{i,7})$$

Put a model on $\mathbf{p_i}$:

- Transform to ensure probabilities sum to 1
- Model can include effects/adjustments for different things e.g. region, data quality, temporal changes, subnational adjustments...
- This is a (Bayesian) hierarchical model. We will learn about these!

More info, see paper:
https://onlinelibrary.wiley.com/doi/10.1002/sim.10199

Maternal mortality summary

- Real world problem, working with WHO and statisticians, epidemiologists, clinicians, public health officials
- So many data problems
- Data complexities lead to relatively complex models
- Substantive area knowledge helps to understand data issues
- Results have big impact (policy, $$$): need to be careful, transparent with assumptions, reproducible

Tools

Tools

- R
- Tidyverse
- RMarkdown/Quarto
- git

R

We will be using R in this course. Pros:

- Free
  - reproducibility
  - portability
- Open
  - large community
  - lots of packages
  - lots of help

RStudio:

- IDE for R that makes using R a lot nicer and easier
- If you haven't already got it, download the free version here: https://posit.co/download/rstudio-desktop/

Tidyverse

- R Packages contain R functions, the documentation that describes how to use them, and sample data.
- The ‘tidyverse’ is “an opinionated collection of R packages designed for data science. All packages share an underlying design philosophy, grammar, and data structures.” https://www.tidyverse.org/
- `ggplot` probably the most well known
- Style of coding fundamentally different to base R.
- A lot of other packages now produce output objects in the ‘tidy’ form

RMarkdown

- Markdown is plain text formatting syntax that can be converted into lots of different outputs (eg HTML, PDF)
- R Markdown allows you to combine Markdown (for the report writing) and embedded R chunks, which are dynamically updated when the document is compiled
- R code can be in chunks or inline (e.g the fourth root of π is 0.7853982)
- These slides are written in RMarkdown and knitted to PDF (beamer)

```
## RMarkdown

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- R Markdown allows you to combine Markdown (for the report writing) and embedded R chunks, which are dynamically updated when the document is compiled
- R code can be in chunks or inline (e.g the fourth root of $\pi$ is `r pi*(1/4)`)
- These slides are written in RMarkdown and knitted to PDF (beamer)

\begin{figure}
\includegraphics[width = 0.8\textwidth]{turtles.png}
\end{figure}
```

RMarkdown

- Good reproducibility tool
- Can do most things you can do in LaTeX (writing math is the same)
- You are expected to write up assignments in RMarkdown or Quarto

Quarto

- The newer version of R Markdown
- Mostly the same, a few bits and pieces improved
- You can use either R Markdown or Quarto to do labs / assignments

git

- git is a version control system (think a more complicated Dropbox)
- Designed for software engineers, but useful for all sorts of code
- Useful for both collaborative and solo projects
- GitHub is useful place to host open source projects

GLMs recap

Motivating examples

Outcomes we may be interested in investigating (in relation to other explanatory variables):

- Police stop and frisks in NYC
- Infant deaths in the US
- Who voted for the Liberal party v other party
- Who voted Liberal, Conservatives, LDP
- Concentration of drug at particular times after ingestion

The take-away: none of these are Normal.

General linear models

Let's start with a recap of general linear models. We observe $y_1, y_2, \ldots y_n$ which are realizations of the random variables $Y_1, Y_2, \ldots, Y_n$

In linear models the y_i's have two pieces:

1. A **systematic part**, with the form

$$E(\mathbf{Y}|\mathbf{X}) = \mu = \mathbf{X}\beta$$

2. A **random part**, where errors are assumed to be i.i.d such that $E[\epsilon] = 0$ and $var[\epsilon] = \sigma^2$. We usually further assume that errors are Normal with constant variance σ^2.

Multiple linear regression

One of the most common examples of a general linear model.

Goal: we are trying to measure the association between response/outcome/dependent variable Y_i and one or more explanatory variables/covariates $X_{i,1}, X_{i,2}, \ldots, X_{i,k}$

- The conditional expectation function (CEF) $E(Y_i|X_{i,1}, X_{i,2}, \ldots, X_{i,k})$ describes the expected value (population mean) of Y_i given values of the variables $X_{i,1}, X_{i,2}, \ldots, X_{i,k}$.
- The CEF has two properties that make it a good measure of statistical association:
  - The CEF decomposition property
  - The CEF ANOVA property

The CEF decomposition property

- Y_i can be decomposed into its CEF and an error term as follows:
$$Y_i = E\left(Y_i \mid X_{i,1}, X_{i,2}, \ldots, X_{i,k}\right) + \varepsilon_i$$
- Note that the residual $\varepsilon_i = Y_i - E\left(Y_i \mid X_{i,1}, X_{i,2}, \ldots, X_{i,k}\right)$ is just the difference between the observed values Y_i and its expected value, or population mean, given $X_{i,1}, X_{i,2}, \ldots, X_{i,k}$.
- One way to interpret the CEF decomposition property is that Y_i can be decomposed into two independent components: a component 'explained by X' and a component 'unexplained by X'.

The CEF ANOVA property

The variance of Y_i can be decomposed into the variance of the CEF and the variance of an error term as follows:

$$\begin{aligned}\mathrm{Var}(Y_i) &= \mathrm{Var}(E(Y_i \mid X_{i,1}, X_{i,2}, \ldots, X_{i,k})) + \mathrm{Var}(\varepsilon_i) \\ &= \mathrm{Var}(E(Y_i \mid X_{i,1}, X_{i,2}, \ldots, X_{i,k})) + E(\mathrm{Var}(Y_i \mid X_{i,1}, X_{i,2}, \ldots, X_{i,k}))\end{aligned}$$

One way to interpret the CEF ANOVA property is that the variance of Y_i can be decomposed into two components: a component 'explained by X' and a component 'unexplained by X'.

Multiple linear regression

MLR is a model for the CEF:

$$\begin{aligned} Y_i &= E\left(Y_i \mid X_{i,1}, X_{i,2}, \ldots, X_{i,k}\right) + \varepsilon_i \\ &= \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_k X_{ik} + \varepsilon_i \end{aligned}$$

Specifically, the most basic MLR model is a simple linear function of the X's and associated parameters β.

An aside: what can this model tell us?

$$
\begin{aligned}
Y_i &= E\left(Y_i \mid X_{i,1}, X_{i,2}\right) + \varepsilon_i \\
&= \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i
\end{aligned}
$$

The MLR assumptions

1. no model misspecification
2. there is independent variation in all of the explanatory variables
   - In other words, none of the explanatory variables are constants, and there are no perfect linear relationships among the explanatory variables
   - e.g. can't have $X_{i1} = X_{i2} + X_{i3}$
3. All variables are from a simple random sample
   - This assumption implies that all members of a population have an equal probability of selection, that all possible samples of size n have an equal probability of selection, and that each observation is independent of all the others

The MLR assumptions

4. The variance of $\varepsilon_i = Y_i - E(Y_i \mid X_{i1}, X_{i2}, \ldots, X_{ik})$ is the same across all values of the explanatory variables
   i.e. $\text{Var}(\varepsilon_i \mid X_{i1}, X_{i2}, \ldots, X_{ik}) = \sigma^2$
   - This is called homoskedasticity
5. The normality assumption $\varepsilon_i = Y_i - E(Y_i \mid X_{i1}, X_{i2}, \ldots, X_{ik})$ is normally distributed

Assumption 1-4 are Gauss-Markov assumptions

Estimation

Minimizing the sum of squared residuals

$$S(\beta) = \sum_{i=1}^{n} \left(y_i - x_i^{\mathrm{T}} \beta \right)^2 = (y - X\beta)^{\mathrm{T}} (y - X\beta)$$

Leads to the MLR-OLS estimator

$$\hat{\beta} = \left(X^{\mathrm{T}} X \right)^{-1} X^{\mathrm{T}} y$$

Sampling distribution of the MLR-OLS estimator

- Under the Gauss Markov and normality assumptions, the OLS estimator, $\hat{\beta}_k$ is normally distributed with a mean equal to

$$E\left(\hat{\beta}_k\right) = \beta_k$$

and variance

$$\text{Var}\left(\hat{\beta}_k\right) = \frac{\sigma^2}{\sum_i \left(X_{ik} - \bar{X}_{ik}\right)^2 (1 - R_k^2)}$$

We can use this property for inference: The sampling distribution of standard error standardized estimator follows a t-distribution with $n - (k+1)$ degrees of freedom.

General linear models

$$
\begin{aligned}
Y_i &\sim N(\mu_i, \sigma^2) \\
\mu_i &= \mathbf{X}_i^T \beta
\end{aligned}
$$

General linear models are not appropriate when

- The range of Y is restricted
- The variance of Y depends on the mean

Generalized Linear Models extend the classical set-up to allow for a wider range of distributions. Introduced by Nelder and Wedderburn (1972) [Later, GAMs in 1990].

Generalized linear models

Generalized linear models

GLMs have an additional piece on top of the classical linear models:

1. **random component**: $Y_i \sim$ some distribution with $E[Y_i|\mathbf{X}_i] = \mu_i$
2. **systematic component**: $\mathbf{X}_i^T\beta$
3. The **link function** that links the random and systematic components $g(u_i) = \mathbf{X}_i^T\beta$

- Set-up is almost the same, particularly in terms of specifying a good linear predictor $\mathbf{X}_i^T\beta$
- Just need to think about the link and the distribution of the outcome

GLMs

$$\begin{aligned} Y_i &\sim G(\mu_i, \phi) \\ E[Y_i|\mathbf{X}_i] &= \mu_i \\ g(\mu_i) &= \mathbf{X}_i^T \beta \end{aligned}$$

- ϕ is the scale parameter.

What can Y be distributed as? In principle, anything. In practice (and original formulation), distributions come from the **exponential family**.

Exponential Family

Exponential Family

The random variable Y belongs to the exponential family of distributions if its support does not depend upon any unknown parameters and its density or probability mass function takes the form

$$p(y|\theta,\phi) = \exp\left(\frac{y\theta - b(\theta)}{\phi} + c(y,\phi)\right)$$

- $\theta = h(\mu)$ depends on the expected value of y and is the **canonical parameter**
- ϕ is the scale parameter (if known: one-parameter family)
- b and c are arbitrary functions

Example: Poisson distribution

$$p(y|\theta, \phi) = \exp\left(\frac{y\theta - b(\theta)}{\phi} + c(y, \phi)\right)$$

Poisson:

$$p(y|\mu) = \frac{\mu^y e^{-\mu}}{y!}$$

Write as

$$p(y|\mu) = \exp\{y \log \mu - \mu - \log y!\}$$

- $\theta = \log \mu$
- $b(\theta) = e^\theta$
- $c(y, \phi) = -\log y!$
- Note that the scale parameter $\phi = 1$ so the variance is entirely determined by the mean

Example: Normal distribution

$$p(y|\theta,\phi) = \exp\left(\frac{y\theta - b(\theta)}{\phi} + c(y,\phi)\right)$$

Normal:

$$p(y|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(y-\mu)^2}{2\sigma^2}\right\}$$

Write as

$$p(y|\mu,\sigma^2) = \exp\left\{\frac{y\mu - \frac{1}{2}\mu^2}{\sigma^2} - \frac{1}{2}\left[\frac{y^2}{\sigma^2} + \log(2\pi\sigma^2)\right]\right\}$$

- $\theta = \mu$
- $b(\theta) = \frac{1}{2}\theta^2$
- $\phi = \sigma^2$
- $c(y,\phi) = -\frac{1}{2}\left[\frac{y^2}{\sigma^2} + \log(2\pi\sigma^2)\right]$

Other examples

Other common examples:

- Binomial
- Gamma
- Negative binomial
- Inverse Gaussian

Lab

Lab this week

- Setting up git, with a small exercise
- Intro to tidyverse and ggplot

Git

- Git is a version control system
- The system tracks changes you make to git repositories ('repos')
- Think of repos as folders
- In order for file versions to be tracked, they need to be **committed** to the git repo
- Think of committing as like saving, but with slightly more steps

GitHub

https://github.com/

- A hosting service for git repos
- You can sign up for free, and host an unlimited number of public or private repos
- You will be submitting lab exercises via GitHub, so you need to set up an account!

The simplest Git/GitHub workflow

- New repo on GitHub
- Clone onto local computer
- Do work on local computer
- Save
- Add and commit to git repo
- Push to GitHub (this means your new work will appear on the GitHub website)

If you are working on your own, on one computer, this is it!

Git/GitHub

- If you are working on a couple of different computers / servers, you may also need to **pull** from GitHub to update any new work done elsewhere
- Git is designed for collaborative work. More complicated workflows have branches, pull requests, merges (more later)

Steps on GitHub

Monica to now demonstrate

- creating a new repository
- Add a collaborator

Then using the terminal (or GitHub Desktop?):

- cloning to your computer
- doing some work
- git status
- add, commit, push

Disclaimer: I use the terminal. You are welcome to use the GitHub Desktop: https://desktop.github.com/ But do not use direct file upload.

This week's lab assessment (git part)

- Make a repo and add me as a collaborator
- Add a text file with your name, program (e.g. statistics Masters), and your favorite thing to do in Toronto
- Push changes to GitHub